Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 5 · SLO 1 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 5 : SLO â€“ 1 | Singly Linked List â€” Traversal and Searching

I. Algorithm for Traversing a Linked List

- Step 1: [INITIALIZE] SET PTR = START
- Step 2: Repeat Steps 3 and 4 while PTR != NULL
- Step 3: Apply Process to PTR -> DATA
- Step 4: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 5: EXIT

II. Algorithm to Print the Number of Nodes

- Step 1: [INITIALIZE] SET COUNT = 0
- Step 2: [INITIALIZE] SET PTR = START
- Step 3: Repeat Steps 4 and 5 while PTR != NULL
- Step 4: SET COUNT = COUNT + 1
- Step 5: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 6: Write COUNT
- Step 7: EXIT

III. Algorithm to Search a Linked List

- Step 1: [INITIALIZE] SET PTR = START
- Step 2: Repeat Step 3 while PTR != NULL
- Step 3: IF VAL = PTR -> DATA
- SET POS = PTR
- Go To Step 5
- ELSE
- SET PTR = PTR -> NEXT
- [END OF IF]
- [END OF LOOP]
- Step 4: SET POS = NULL
- Step 5: EXIT